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Creating a Realistic Smoke Effect in Blender

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1. Introduction

This laboratory work demonstrates how to create a realistic smoke effect using **Blender**. The exercise covers essential 3D techniques including modeling, applying modifiers, configuring smoke simulations, setting up lighting and materials, and preparing final render outputs. The goal is to understand how different components—geometry, physics simulation, shaders, and render settings—combine to produce convincing visual effects. This lab also highlights practical challenges such as rendering time, hardware limitations, and scene optimization.

2. Scene Preparation

2.1 Starting the Project

1. Open Blender.
2. Select all objects and delete them to start with a clean scene.

3. Modeling the Smoke Emission Setup

3.1 Creating the Cylinder

The cylinder acts as part of the smoke environment.

- Add a **Cylinder** and adjust its parameters (radius, depth, vertices).
- With the cylinder selected:
 - Go to **Modifiers** → **Generate** → **Solidify**
 - Adjust the *thickness* to shape the object.
 - Add another modifier: **Generate** → **Bevel**
 - Adjust bevel width and segments for smooth edges.
 - Right-click → **Shade Smooth** to smooth the surface.

3.2 Adding the Ground Plane

- Add a **Plane**, scale it, and position it under the cylinder.

3.3 Camera Setup

- Add a **Camera**.
- Move, rotate, and position it to frame the scene correctly.

3.4 Additional Plane Panels

These help create controlled light or shadow effects.

- Add another **Plane**, rotate and move it.
- Duplicate it as needed.

4. Smoke Simulation Setup

4.1 Creating the Smoke Emitter

- Add a **UV Sphere** and adjust its parameters and scale.
- This sphere will emit the smoke.

4.2 Setting the Cylinder as a Collider

- Select the cylinder → **Physics** → **Fluid**
- Under *Type*, choose **Effector** to allow smoke interaction with the geometry.

4.3 Creating the Smoke Domain

- Select the sphere → **Object** → **Quick Effects** → **Quick Smoke**
Blender automatically creates:
 - *A Smoke Domain* (cube)
 - Smoke settings for the sphere (emitter)
- Move or resize the smoke domain as needed.

4.4 Adjusting Domain Resolution

To improve smoke detail:

- Select the smoke domain → **Physics Properties**
 - Increase **Resolution** to **128**.
 - Enable **Adaptive Domain**.
 - Under **Gas**, adjust **Buoyancy Density**.
 - Enable **Noise** and set **Type** = **All** for higher realism.

4.5 Adding Turbulence Force Field

- Add → **Force Field** → **Turbulence**
- Increase **Strength** to intensify smoke movement.

4.6 Adjusting Emitter Properties

- Select the sphere:
 - Increase **Sampling Substeps** for smoother smoke emission.
 - Change smoke color to **white**.

4.7 Baking the Simulation

- Select smoke domain → **Physics** → **Bake All**
- This calculates the full smoke animation for better final results.

5. Material and Shading Setup

5.1 Stopping the Simulation

Set the timeline to frame **100** to preview the smoke at a stable moment.

5.2 Smoke Shader Adjustments

- Select the smoke domain.
- Switch to **Shading Workspace**.
- Add:
 - **Attribute Node**
 - **Vector Math Node**
- These provide color and depth variation for a more realistic smoke effect.

6. Lighting Setup

6.1 Adding Light Sources

- Add **Area Lights**.
 - Adjust:
 - Color
 - Position
 - Power
- These help highlight the smoke and enhance realism.

7. Floor Material Setup

7.1 Texturing the Floor

- Select the floor plane.
- Create a **New Material**.
- Add an **Image Texture**:
 - Use **Ctrl + T** (Node Wrangler) to generate Texture Coordinate & Mapping nodes.
 - Select a floor image.
- Add **ColorRamp** for contrast adjustments.
- Add a **Bump Node** for surface depth.

8. Glow Effect

8.1 Creating Glow Objects

- Add a new **Cylinder**.
- Adjust parameters to form a glowing object (like a lamp or neon tube).
- Apply emission material for glow inside the scene.

9. Final Rendering

9.1 Render Settings

- Go to **Render Properties**:
 - Engine: **Cycles**
 - Device: **CPU** (required due to Intel UHD / Iris Xe)
 - Samples: **64**
 - Enable **Denoising (Intel OpenImageDenoise)**

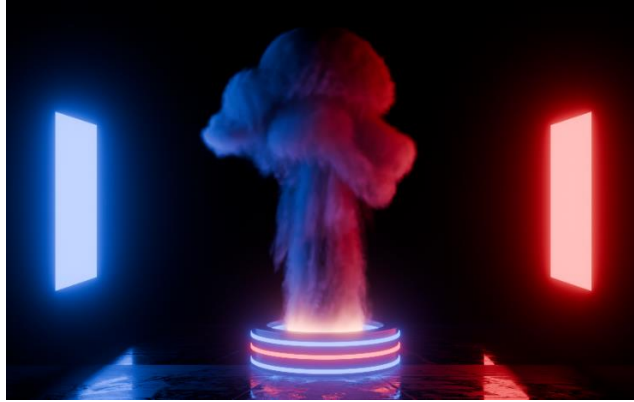
9.2 Output Settings

- Start Frame: **1**
- End Frame: **250**

9.3 Rendering

Render Image

- Use **Render → Render Image** for still images.



Render Animation

- File Format: **FFmpeg Video**
- Container: **MPEG-4**
→ Produces a single **MP4** video file.

10. Problems Encountered

10.1 Extremely Long Rendering Times

Rendering was significantly slow due to:

- **High number of frames (250)**: More frames require more processing.
- **Smoke simulation complexity**: High domain resolution + noise + turbulence increase computation.
- **Using CPU Rendering**:
The laptop uses **Intel UHD / Iris Xe Graphics**, which:
 - Does **not** support CUDA
 - Has limited OpenCL support
 - Cannot use GPU acceleration in Cycles
→ Blender must use **CPU**, which is much slower.

These limitations resulted in long simulation bakes and render times.

11. Conclusion

This lab demonstrated a complete workflow for creating realistic smoke simulations in Blender including modeling, physics configuration, shading, lighting, and rendering. Although the final effect is convincing, rendering performance is heavily constrained by hardware limitations. Working with adaptive domain, lower resolutions, fewer frames, and optimized materials can

improve rendering speed.

Overall, the exercise provides valuable experience with Blender's physics simulation tools and highlights the importance of balancing visual quality with computing resources.